# Diagonalization



Recall from last class:

- If $A$ is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of $A$ if there exists a non-zero vector $v$, called an **eigenvector** corresponding to $\lambda$, such that
$$Av = \lambda v.$$
- If $\lambda$ is an eigenvalue of an $n \times n$ matrix $A$, then the set of all solutions $v$ to $Av = \lambda v$ is the **eigenspace** of $A$ corresponding to $\lambda$.
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \ldots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_p$ of an $n \times n$ matrix $A$, then $v_1, \ldots, v_p$ are linearly independent.
- If $A$ is an $n \times n$ matrix, then the characteristic polynomial of $A$, denoted by $p_A(\lambda)$, is given by
$$p_A(\lambda) = \det(A - \lambda I_n).$$
- $\lambda$ is an eigenvalue of an $n \times n$ matrix $A$ if and only if $\lambda$ is a zero of the characteristic polynomial.
- If $A$ and $B$ are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).

**1.** Let $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$. Find eigenvalues and eigenspaces of $A$.

**Definition 1.** *An $n \times n$ matrix $A$ is* **diagonalizable** *if $A$ is similar to a diagonal matrix.*

**Diagonalization Theorem:** An $n \times n$ matrix $A$ is diagonalizable if and only if $A$ has $n$ linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with $D$ diagonal if and only if the columns of $P$ are $n$ linearly independent eigenvectors of $A$. In this case the diagonal entires of $D$ are the corresponding eigenvalues of $A$.

**Definition 2.** *A basis of* $\mathbb{R}^n$ *diagonalizing* $A$ *is called an* **eigenbasis**.

**2.** Prove the Diagonalization Theorem.

**3.** Let $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ If possible, diagonalize $A$.

**4.** Let $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ If possible, diagonalize $A$.

**Theorem 6:** An $n \times n$ matrix $A$ with $n$ distinct eigenvalues is diagonalizable.

**5.** Prove the theorem.

**6.** Let $T\colon \mathbb{P}_3 \to M_{2\times 2}$ be given by

$$T(a_0 + a_1 t + a_2 t^2 + a_3 t^3) = \begin{bmatrix} a_0 - a_1 & a_2 - a_3 \\ a_1 - a_0 & a_3 - a_2 \end{bmatrix}.$$

Find bases for the kernel and range of $L$. Justify your answer. Find the standard matrix of $L$ relative to the standard bases.

**7.** Let $U_{2\times 2}$ be the space of upper triangular 2 by 2 matrices. Let $T : \mathbb{P}_2 \to U_{2\times 2}$ be a linear transformation given by

$$T(f) = \begin{bmatrix} f(0) & f'(1) \\ 0 & f''(2) \end{bmatrix}.$$

Find the matrix of $T$ relative to the standard bases. Find the eigenvalues and eigenspaces of the matrix of $T$ relative to the standard bases.

**8.** Let $T$ be the linear transformation $\mathbb{R}^3 \to \mathbb{R}^3$ with matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

(a) Compute the dimension of the kernel $T$ and a basis for the kernel.

(b) Compute the rank of $A$ and a basis for the image of $T$.

**9.** Let

$$A = \begin{bmatrix} -10 & -18 \\ 9 & 17 \end{bmatrix}.$$

(a) Compute the eigenvalues of $A$.

(b) Compute a basis for each eigenspace of $A$.

(c) Compute a matrix $B$ such that $B^3 = A$.

**10.** Let $T : \mathbb{P}_2 \to \mathbb{R}^3$ be defined by

$$T(p(t)) = \begin{bmatrix} p(1) \\ p(2) \\ p(3) \end{bmatrix}$$

for every polynomial $p(t) = a_2 t^2 + a_1 t + a_0$ in $\mathbb{P}_2$.

(a) Show that $T$ is a linear transformation.

(b) Find a matrix $M$ such that $T(p(t)) = M[p(t)]_{\mathcal{B}}$ where $\mathcal{B} = \{1, t, t^2\}$.

(c) Show that for all $c_1, c_2, c_3 \in \mathbb{R}$ there is a polynomial $p(t) \in \mathbb{P}_2$ such that $p(1) = c_1$, $p(2) = c_2$, and $p(3) = c_3$.

**11.** Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be a basis for $\mathbb{R}^3$. Suppose that $A$ and $B$ are $3 \times 3$ matrices such that $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are eigenvectors of both $A$ and $B$. Show that $AB = BA$.

**12.** Let $A$ be an $n \times n$ matrix which is not invertible, and define $K_r$ to be the nullspace of $A^r$. That is, $K_1 = \text{Nul}(A)$, $K_2 = \text{Nul}(A^2)$, etc.

(a) Prove that, for each $j = 1, 2, 3, \ldots$, $K_j$ is a subspace of $K_{j+1}$, so that

$$K_1 \subseteq K_2 \subseteq K_3 \subseteq K_4 \subseteq \cdots .$$

(b) Prove that, if for some $r$ we have $K_r = K_{r+1}$, then $K_r = K_n$ for all $n \geq r$.

**13.** True or False Questions.

(a) There is a $2 \times 4$ matrix $A$ with rank 1 and

$$\text{Nul}(A) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

(b) Let $V$ and $W$ be vector spaces and $T : V \to W$ be a one-to-one linear transformation. If $\{T(v_1), \cdots, T(v_k)\}$ is a basis for $W$, then $\{v_1, \cdots, v_k\}$ is a basis for $V$.

(c) The set $\{A \in M_{n\times n}(\mathbb{R}) : A \text{ is invertible. } \}$ is a subspace of $M_{n\times n}(\mathbb{R})$.

(d) Let $A$ be an $n$ by $n$ matrix such that the sum of each row is equal to 1. Then $A - I$ is not invertible.

(e) Let $E_1$ and $E_2$ be eigenspaces of distinct eigenvlaues of a matrix $A$. Then $E_1 \cap E_2 = \{0\}$.

(f) Let $W_1$ and $W_2$ be 3-dimensional subspaces of $\mathbb{R}^5$, then $W_1$ and $W_2$ have a non-zero vector in common.

(g) $V$, $W$ subspaces of $\mathbb{R}^n$, then $V \cup W$ is a subspace.

(h) Let $W_1$ and $W_2$ be subspaces of $\mathbb{R}^n$ such that $\dim(W_1) \leq \dim(W_2)$. Then $W_1 \subset W_2$.

(i) If $x$ is an eigenvector of a matrix $A$, then $x$ is also an eigenvector of $A^3$.

(j) If $A$ is an $n \times n$ matrix and $x, y$ are two eigenvectors of $A$, then $x+y$ is also an eigenvector of $A$.

(k) If $V$ and $W$ are vector spaces of dimension $n$ and $T : V \to W$ is a one-to-one linear transformation, then $T$ is onto.